Toward a Proof of the Yang–Mills Existence and Mass Gap Problem – The Wilson Lattice Measure in Lean 4: Established Results and Open Obligations

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Abstract

This paper does not prove the Yang–Mills existence and mass gap conjecture, and it does not prove a positive mass gap for any quantum Yang–Mills theory. It is the record of an attempt. We report what a machine-checked campaign on the four-dimensional SU(2) Wilson lattice measure established, which obligations it left open, and why the automated search was stopped on 9 November 2026: it had drifted into reward hacking, producing theorems that satisfied its checks without advancing the problem. We publish the whole Lean 4 corpus of the campaign, 55 modules and 105,960 lines, compiled from a clean copy with zero `sorry` declarations and only Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`). A kernel check certifies each stated implication and says nothing about whether a statement means what its name suggests, so we separate the corpus into results we stand behind and material published as it is, and we document the failure patterns with named examples. The results we stand behind are exact finite-volume statements about the actual Wilson measure (gauge and translation invariance, Osterwalder–Seiler reflection positivity, a two-sided volume law for the Haar mass of low-action configurations, and a volume-uniform action-density bound), weak compactness of the finite-volume laws, an abstract resolvent theorem for self-adjoint generators, and negative controls that refute tempting shortcuts on explicit models. The open part is every step that the conjecture actually needs: uniform multiscale control, a unique continuum limit, full Euclidean covariance, clustering, nontriviality, and a positive physical decay rate. We list those obligations as numbered Problems and say for each what blocked us. Section 6 describes how the automated search went wrong and what now guards against it.

Key words. Yang–Mills theory, lattice gauge theory, Wilson action, reflection positivity, mass gap, formal verification, Lean 4.
AMS subject classifications. 81T13, 81T25, 81T08, 60B10, 68V15.

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What is actually proved

Every row is a declaration in the published library. "Wilson measure" means the probability measure with density $e^{-\beta S}/Z$ against normalized product Haar measure on the links of the periodic lattice $(\mathbb{Z}/L)^4$, with S the SU(2) plaquette action. None of the rows is a statement about a continuum quantum field theory.

Result	Hypotheses	Conclusion	Kind
2.1.1	finite L, any β	the Wilson action is invariant under every lattice gauge transformation	exact, finite volume
2.2.1	$\beta \geq 0$, $\delta > 0$	$P_\beta(S \geq \varepsilon) \leq e^{-\beta(\varepsilon-\delta)}/c(L,\delta)$ with c the Haar mass of $\{S < \delta\}$	exact, finite volume
2.2.2	$0 < \delta \leq 96L^4$; and $L \geq 2$, $4\delta \leq m(L)$	two-sided bounds on $c(L,\delta)$; $\log(1/c)$ is of order $L^4 \log L$	exact, finite volume

Result	Hypotheses	Conclusion	Kind
2.2.3	$0 < \delta < \varepsilon$, $\delta \le 96$, one fixed β above an explicit threshold	$P_\beta(S \ge \varepsilon L^4)$ is bounded uniformly in L and tends to 0 as $L \to \infty$	exact, uniform in volume
2.2.4	$\varepsilon > 0$, fixed $\beta \ge 0$	$P_\beta(S \ge \varepsilon) \to 1$ as $L \to \infty$	exact; limits 2.2.3
2.3.1	any sides L_n, couplings $\beta_n \ge 0$	a subsequence of the periodically extended laws converges weakly; the limit is invariant under global gauge rotations and integer translations	exact; compactness
2.4.1	$L = 2M$, $\beta \ge 0$, H continuous and supported on the closed positive half	$\langle (H \circ \Theta)\, H \rangle_\beta \ge 0$	exact, finite volume
2.5.1	$L \ge 2$, $\beta \ge 0$, $0 < t \le \frac{1}{2}$	$P_\beta(s(U_p) \ge t) \ge 1/(1 + e^{24\beta t})$ for every plaquette	exact, uniform in volume
2.5.2	as 2.5.1, P a separated family of plaquettes	$P_\beta(\text{all } s(U_p) < t) \le \exp(-\lvert P\rvert/(1 + e^{24\beta t}))$	exact, uniform in volume
2.5.3	$\beta = b \log k$, threshold $2\varepsilon/\sqrt{\beta}$	the event that k^3 face plaquettes are all small has probability tending to 0, uniformly in $L \ge 2k$	exact, uniform in volume
2.6.1	flat background, any L	the forest-slice curl form is bounded below, and its inverse decays geometrically in graph distance	exact, linear algebra
2.6.2	weight $e^{t\cos\theta}\sin^2\theta$ on $(0,\pi)$, any real t	Poincaré inequality with constant 3	exact, one variable
3.1.1	A self-adjoint, $\langle u, Au\rangle \ge m\lVert u\rVert^2$ on its domain, $E < m$	$A - E$ has a bounded everywhere-defined inverse of norm at most $1/(m - E)$	abstract Hilbert space
3.1.2	A self-adjoint generator of $T(t)$ with $\lVert T(t)u\rVert \le e^{-mt}\lVert u\rVert$ on its domain	same resolvent conclusion	abstract Hilbert space
4.1.1–4.1.4	none	four false strengthenings are refuted on explicit models	negative control

Results 3.1.1 and 3.1.2 contain no lattice object. They become statements about Yang–Mills only after the physical Hilbert space, its generator and the decay estimate have been constructed, and none of those has been.

Introduction

Purpose. We present the verified residue of an attempt on the following problem of Jaffe and Witten [2]: for every compact simple gauge group G, prove that a nontrivial quantum Yang–Mills theory exists on $\mathbb{R}^4$, satisfying the axioms of Osterwalder–Schrader or Wightman, and that its Hamiltonian has spectrum contained in $\{0\} \cup [\Delta, \infty)$ for some $\Delta > 0$. We do not solve it. The purpose of the paper is to make the attempt checkable: every claim that we keep is a kernel-checked Lean 4 declaration that anyone can rebuild, and every claim that we could not keep is stated as an open Problem.

History. Wilson [8] introduced the lattice regularization used here. Osterwalder and Seiler [7] proved reflection positivity of the lattice measure and exponential clustering at strong coupling, where polymer expansions of the kind analysed by Kotecký and Preiss [3] converge. Bałaban [1] controlled the ultraviolet problem by a multiscale renormalization group analysis on a

finite-volume torus. The reconstruction of a quantum theory from Euclidean correlation functions is due to Osterwalder and Schrader [6]. No argument is known that joins the weak-coupling ultraviolet regime to a positive physical mass in the continuum.

Basic idea of the program. Fix the SU(2) Wilson measure on a periodic lattice, prove exact statements about that measure in a proof assistant, take limits only through theorems whose hypotheses name what is still missing, and attach to every hypothesis a negative control showing that it does not follow from the others.

Main parts in more detail. The program has six parts. (i) *Uniform weak-coupling control*: bounds on the block-averaged Wilson integral that are uniform in volume. Section 2 serves it; its status is Problem 5.1.1. (ii) *A unique continuum limit* of renormalized Wilson loops. Theorem 2.3.1 gives subsequential limits only; see Problem 5.1.2. (iii) *Euclidean covariance and reflection positivity* of the limit. Theorem 2.4.1 gives positivity at finite volume; see Problem 5.1.3. (iv) *Clustering and nontriviality*; see Problem 5.1.4. (v) *A positive physical decay rate* and the spectral gap of the reconstructed Hamiltonian. Section 3 supplies the abstract last step; see Problem 5.1.5. (vi) *All compact simple groups*; see Problem 5.1.6.

Main results. Theorems 2.1.1–2.5.3 are exact statements about the finite-volume Wilson measure, and Theorems 2.6.1 and 2.6.2 are exact auxiliary estimates. Theorems 3.1.1 and 3.1.2 are abstract. Propositions 4.1.1–4.1.4 are negative controls.

What is new, and what was substituted. Gauge invariance, reflection positivity and weak compactness are known mathematics; the contribution is their formal verification for one fixed definition of the measure. The explicit two-sided volume law of Theorem 2.2.2 and the explicit threshold of Theorem 2.2.4 were derived for this library, as were the plaquette tail bounds of Theorems 2.5.1–2.5.3; we have not found the constants in the literature and make no claim that the orders of growth are new. The resolvent theorems are standard functional analysis. Nothing here re-derives a theorem of Bałaban, of Osterwalder and Seiler beyond the stated positivity, or of Kotecký and Preiss.

Audience and self-containment. The paper is for readers who want to check the claims rather than trust them. The archive contains the Lean sources, the project files and a build log. Pins: toolchain `leanprover/lean4:v4.34.1`; Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`; source revision `0ec68a0f37923677f56a6124412951840850d9b0`.

Acknowledgments. The library depends on Mathlib [5] and Lean 4 [4].

1 The model

In this chapter, we fix the objects that every later statement refers to.

1.1 Lattice, links and action

Definition 1.1.1 (Jesse; [8])**.** Let $L \geq 1$. A *gauge configuration* assigns an element $U_e \in \mathrm{SU}(2)$ to every link e of the periodic lattice $(\mathbb{Z}/L)^4$. For a plaquette p let U_p be the ordered product of the four link variables around p, and put $s(U_p) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p$. The *Wilson action* at coupling β is

$S_\beta(U) = \beta \sum_p s(U_p)$, and $S = S_1$. The *Wilson measure* is the probability measure $Z_\beta^{-1} e^{-S_\beta}\, d\mu$ with μ the normalized product Haar measure. In Lean these are `wilsonAction`, `gaugeHaarMeasure`, `wilsonMeasure` and `wilsonProbability`.

2 Exact results for the finite-volume Wilson measure

In this chapter, we state the results that hold for the measure of Definition 1.1.1 with no further hypothesis.

2.1 Gauge invariance

Theorem 2.1.1 (Jesse; [8])**.** *For every L, every β and every lattice gauge transformation g, $S_\beta(g \cdot U) = S_\beta(U)$.*

Proof. A gauge transformation conjugates each plaquette holonomy by the group element at its base site, and the trace is invariant under conjugation. *Formal verification:* `wilsonAction_gaugeTransform` in `JesseMath/Physics/LatticeYangMills.lean`; kernel accepted; axioms `propext`, `Classical.choice`, `Quot.sound`; source digest in Appendix A. □

2.2 Action tails and the volume law

Write P_β for the Wilson measure on the lattice of side L and $c(L, \delta) = \mu\{S < \delta\}$.

Theorem 2.2.1 (Jesse)**.** *If $\beta \geq 0$ and $\delta > 0$ then, for every real ε,*

$$P_\beta(S \geq \varepsilon) \;\leq\; \frac{e^{-\beta(\varepsilon - \delta)}}{c(L, \delta)}. \tag{2.2.1}$$

Proof. Bound the numerator by $e^{-\beta\varepsilon}$ and the partition function from below by its integral over $\{S < \delta\}$. *Formal verification:* `haarWilsonMeasure_action_tail` in `JesseMath/Physics/YangMillsWeakCoupling.lean`; kernel accepted; standard axioms. □

Theorem 2.2.2 (Jesse)**.** *Let $N = 4L^4$ be the number of links and $m(L) = L^3 \lfloor L/2 \rfloor$. If $0 < \delta \leq 96 L^4$ then*

$$\Big(\frac{\delta}{2400\, L^4}\Big)^{2N} \;\leq\; c(L, \delta), \tag{2.2.2}$$

and if $L \geq 2$ and $4\delta \leq m(L)$ then

$$\log c(L, \delta) \;\leq\; m(L) \log 2 - \frac{m(L)}{4} \log \frac{m(L)}{4\delta}. \tag{2.2.3}$$

Proof. For (2.2.2), cover SU(2) by translates of a small neighbourhood of the identity to bound the single-link mass from below, and note that four links close to the identity give a plaquette of small action. For (2.2.3), choose $m(L)$ links each lying in exactly one of a family of disjoint plaquettes, integrate over those links first, and apply a union bound. *Formal verification:* `haarVacuumSublevelMass_ge` and `log_haarVacuumSublevelMass_le` in `JesseMath/Physics/YangMillsUniformHaarNeighbourhood.lean`; kernel accepted; standard axioms. □

Remark 2.2.3. By (2.2.2) and (2.2.3) the entropy cost $\log(1/c(L, \delta))$ at a fixed threshold δ is of order $L^4 \log L$. Hence the right side of (2.2.1) cannot be made small uniformly in L at fixed β and fixed δ.

Theorem 2.2.4 (Jesse)**.** *Let $0 < \delta < \varepsilon$ with $\delta \le 96$ and let $\beta > 8\log(2400/\delta)/(\varepsilon - \delta)$. Then for every $L \ge 1$*

$$P_\beta(S \ge \varepsilon L^4) \;\le\; \exp\bigl(-(\beta(\varepsilon - \delta) - 8\log(2400/\delta))\bigr), \tag{2.2.4}$$

and the left side tends to 0 as $L \to \infty$ at this fixed β.

Proof. Apply Theorem 2.2.1 at the threshold δL^4, where the lower volume law has a base that does not depend on L. *Formal verification:* `haarWilsonMeasure_density_tail_uniform_explicit` in `JesseMath/Physics/YangMillsUniformHaarNeighbourhood.lean`; kernel accepted; standard axioms. □

Theorem 2.2.5 (Jesse)**.** *For every $\varepsilon > 0$ and every fixed $\beta \ge 0$, $P_\beta(S \ge \varepsilon) \to 1$ as $L \to \infty$.*

Proof. A lower bound on $\log Z_\beta$ of order $-L^4 \log(\beta + 1)$ shows that the sublevel $\{S < \varepsilon\}$ has vanishing probability. *Formal verification:* `haarWilsonMeasure_action_tail_tendsto_one_fixed_coupling`, same module; kernel accepted; standard axioms. □

Remark 2.2.6 (Scope)**.** Theorem 2.2.4 controls the average plaquette energy. It gives no information on correlations, on a renormalization trajectory or on a mass, and Theorem 2.2.5 shows that it cannot be upgraded to a bound on the total action.

2.3 Weak limits and their symmetries

Theorem 2.3.1 (Jesse; Prokhorov)**.** *Let $L_n \ge 1$ and $\beta_n \ge 0$ be arbitrary sequences and extend each finite-volume Wilson measure periodically to the space of $\mathrm{SU}(2)$-valued functions on the links of $\mathbb{Z}^4$. Then some subsequence converges weakly to a probability measure ν, the Wilson expectations of every continuous function converge to its ν-integral along that subsequence, and ν is invariant under every global gauge rotation and every translation in $\mathbb{Z}^4$.*

Proof. The configuration space is a countable product of compact metric spaces, so the set of probability measures is sequentially compact. The two symmetries hold at each finite volume and are implemented by continuous maps, so they pass to weak limits. *Formal verification:* `periodicWilsonProbability_symmetric_subsequence` in `JesseMath/Physics/YangMillsContinuumCompactness.lean`; kernel accepted; standard axioms. □

Remark 2.3.2 (Scope)**.** The limit lives on unscaled lattice links. It is not a continuum field, it need not be unique, and with $L_n = 1$, $\beta_n = 0$ the theorem applies to a constant sequence (`tests/YangMillsSixWaveScopeControl.lean`).

2.4 Reflection positivity

Theorem 2.4.1 (Jesse; [7])**.** *Let $L = 2M$, $\beta \ge 0$ and let Θ be the reflection $x_0 \mapsto -x_0$ acting on configurations. If H is a continuous real function of the configuration that depends only on the links of the closed half $0 \le x_0 \le M$, then*

$$\int H(\Theta U)\, H(U)\, dP_\beta(U) \;\ge\; 0. \tag{2.4.1}$$

Proof. Split the action into the part supported in the positive half and its reflection, and write the integral as the integral over the links of the reflection hyperplanes of a square. *Formal verification:* `wilson_reflection_positive` in `JesseMath/Physics/YangMillsOsterwalderSchraderInheritance.lean`; kernel accepted; standard axioms. □

2.5 Plaquette tails and the small-field event

Theorem 2.5.1 (Jesse)**.** *Let $L \geq 2$, $\beta \geq 0$ and $0 < t \leq \frac{1}{2}$. For every plaquette p,*

$$P_\beta(s(U_p) \geq t) \;\geq\; \frac{1}{1 + e^{24\beta t}}, \tag{2.5.1}$$

and if $\beta \geq \frac{1}{12}$ then the expectation of $s(U_p)$ is at least $1/(24(1+e)\beta)$.

Proof. In SU(2) the element $g + g^{-1}$ is a scalar, so moving one link by g and by g^{-1} changes the action with no first-order term. Convexity of the exponential turns this into a transfer of mass between events. *Formal verification:* `wilson_plaquette_tail` and `wilson_plaquette_expectation_lower` in `JesseMath/Physics/YangMillsWilsonPlaquetteTail.lean`; kernel accepted; standard axioms. □

Theorem 2.5.2 (Jesse)**.** *Under the hypotheses of Theorem 2.5.1, let P be a finite family of plaquettes whose first links avoid each other's boundaries. Then*

$$P_\beta(s(U_p) < t \text{ for all } p \in P) \;\leq\; \exp\Big(-\frac{|P|}{1 + e^{24\beta t}}\Big). \tag{2.5.2}$$

Proof. Apply the mass transfer of Theorem 2.5.1 one plaquette at a time, on an event the moved link cannot see. *Formal verification:* `wilson_smallField_tower` in `JesseMath/Physics/YangMillsWilsonSmallField.lean`; kernel accepted; standard axioms. □

Theorem 2.5.3 (Jesse)**.** *Let $b, \varepsilon, \delta > 0$. There is k_0 such that for all $k \geq k_0$ and all $L \geq 2k$, at coupling $\beta = b \log k$ the event that all k^3 plaquettes of a three-dimensional face have $s(U_p) < 2\varepsilon/\sqrt{\beta}$ has probability at most δ.*

Proof. Insert $t = 2\varepsilon/\sqrt{\beta}$ into (2.5.2); the exponent $k^3/(1 + e^{48\varepsilon\sqrt{b \log k}})$ tends to infinity. *Formal verification:* `wilson_smallField_collapse`, same module; kernel accepted; standard axioms. □

Remark 2.5.4 (Scope)**.** Theorem 2.5.3 is an obstruction, not a step forward: a renormalization argument cannot assume that every plaquette of a block of fixed physical size is small. Large fields must be treated, which is the hard part of Problem 5.1.1.

2.6 Two auxiliary estimates

Theorem 2.6.1 (Jesse)**.** *On the fine lattice $(\mathbb{Z}/2L)^4$ with a tree gauge that leaves 45 free links per cell, the quadratic form $Q(a) = \sum_p (\operatorname{curl} a)_p^2$ on functions of the free links satisfies $\frac{1}{20}\sum a^2 \leq Q(a) \leq 24 \sum a^2$, and the inverse of its matrix has entries bounded by $20\,(479/480)^d$ in the distance d of the conflict graph.*

Proof. The lower bound uses an explicit table of plaquette fillings; decay follows from a Neumann series for a sparse positive matrix. *Formal verification:* `forestHessian_lower`, `forestHessian_upper` and `forestHessian_inverse_decay` in `JesseMath/Physics/YangMillsForestSliceHessian.lean`; kernel accepted; standard axioms. □

Theorem 2.6.2 (Jesse)**.** *Let $w(\theta) = e^{t\cos\theta}\sin^2\theta$ on $(0, \pi)$ with t real. If u is continuously differentiable with $\int u\, w = 0$, then $3\int u^2 w \leq \int u'^2 w$.*

Proof. A dual weighted Hardy argument. *Formal verification:* `radial_poincare` in `JesseMath/Physics/YangMillsWilsonLinkRadialPoincare.lean`; kernel accepted; standard axioms. □

Remark 2.6.3 (Scope)**.** Theorem 2.6.1 concerns the flat background only; it is not shown to be the Hessian of the Wilson action in any gauge. In Theorem 2.6.2 the weight is what the one-link Wilson weight would become in polar coordinates on SU(2); that change of variables is not formalized, so the theorem is a statement about one real variable.

3 An abstract resolvent theorem

In this chapter, we record the functional-analytic step that would turn a decay estimate into a spectral gap. It contains no lattice object.

3.1 Coercivity and decay

Theorem 3.1.1 (Jesse; standard)**.** *Let A be a densely defined self-adjoint operator on a complex Hilbert space with $\mathrm{Re}\langle u, Au\rangle \geq m\|u\|^2$ for all u in its domain, and let $E < m$ be real. Then there is a bounded operator R defined everywhere with $(A - E)Rf = f$ for all f, $R(A - E)u = u$ for all u in the domain, and*

$$\|R\| \leq \frac{1}{m - E}. \tag{3.1.1}$$

Proof. Coercivity gives $\|(A - E)u\| \geq (m - E)\|u\|$, so the range is closed because the graph is complete; self-adjointness shows that the orthogonal complement of the range is zero. *Formal verification:* `selfAdjoint_resolvent_below_coercive_bound` in `JesseMath/Physics/YangMillsPhysicalGap.lean`; kernel accepted; standard axioms. □

Theorem 3.1.2 (Jesse; standard)**.** *Let A be self-adjoint and let $T(t)$ be bounded operators with $T(0)u = u$, right derivative $-Au$ at $t = 0$, and $\|T(t)u\| \leq e^{-mt}\|u\|$ for $t > 0$, for all u in the domain of A. Then the conclusion of Theorem 3.1.1 holds for every real $E < m$.*

Proof. Differentiating $\|T(t)u\|^2 \leq e^{-2mt}\|u\|^2$ at $t = 0$ gives the coercivity hypothesis of Theorem 3.1.1. *Formal verification:* `selfAdjoint_resolvent_of_evolution_decay`, same module; kernel accepted; standard axioms. □

Remark 3.1.3 (Assumptions)**.** Self-adjointness, the domain, the generator identity and the decay estimate are hypotheses. For Yang–Mills each of them is open. The same module contains a declaration, `j14_h2h_six_wave_master_physical_gap_and_compact_simple_group_synthesis`, whose name suggests more than it proves: its hypotheses include the decay rate and it uses no construction from Section 2. We keep it in the archive so that the audit trail is complete, and we do not count it as a result.

4 Negative controls

In this chapter, we list statements that would have shortened the route and are false.

4.1 Four refuted shortcuts

Proposition 4.1.1 (Jesse)**.** *Along the explicit dyadic cutoff path of the library, it is not the case that the joint Wilson-loop laws are eventually time mixing. Hence mixing of a limit cannot be inherited from the cutoff laws.*

Proof. Formal verification: `dyadic_cutoffs_not_eventually_mixing` in `tests/YangMills OsterwalderSchraderInheritanceNegativeControl.lean`; kernel accepted. □

Proposition 4.1.2 (Jesse)**.** *There is a law on the common loop space that is reflection positive, invariant under all the symmetries considered and time mixing, and is trivial.*

Proof. The constant Dirac law. *Formal verification:* `constantLaw_trivial`, same file; kernel accepted. □

Proposition 4.1.3 (Jesse)**.** *Tightness of the joint Wilson-loop laws does not imply their convergence along every cutoff path: an oscillating path has two limit points.*

Proof. Formal verification: `convergence_obligation_not_automatic` in `tests/YangMills RenormalizedObservableLimitsNegativeControl.lean`; kernel accepted. □

Proposition 4.1.4 (Jesse)**.** *Unbounded renormalization factors destroy the uniform tail condition that is equivalent to tightness.*

Proof. Formal verification: `not_tailObligation_growingFactors`, same file; kernel accepted. □

5 The route to the full conjecture

In this chapter, we state what is missing. The library names the missing inputs as fields of one structure, `ObligationsExceptReflection` in `JesseMath/Physics/YangMillsRenormalized ObservableLimits.lean`; every field is an undischarged hypothesis.

5.1 Open obligations

Problem 5.1.1 (Uniform multiscale control)**.** *Requires:* an identity for each block-averaged law and a norm for the whole remainder that stays bounded through a number of renormalization steps that diverges as the lattice spacing tends to zero, uniformly in volume and including large fields. *The library provides:* the single-step fluctuation remainder at fixed volume and Theorems 2.2.1–2.2.5. *Obstruction:* every estimate we proved is either at fixed volume or controls only the action density; a single-link oscillation bound below one does not give a contraction, and a four-link parity example shows the inference fails.

Problem 5.1.2 (Unique continuum limit)**.** *Requires:* tightness of renormalized Wilson loops with the unbounded perimeter and corner factors expected in four dimensions, and agreement of all limit laws. *The library provides:* Theorem 2.3.1 and the equivalence of tightness with a tail condition. *Obstruction:* Propositions 4.1.3 and 4.1.4.

Problem 5.1.3 (Euclidean covariance and positivity of the limit)**.** *Requires:* reflection positivity for the full positive-time algebra of the limit, and invariance under all translations and rotations of $\mathbb{R}^4$. *The library provides:* Theorem 2.4.1 and the symmetries of Theorem 2.3.1. *Obstruction:* positivity at a finite periodic cutoff holds only for observables in a half lattice, so it must be passed to the limit observable by observable, and a regularity hypothesis is then needed; only lattice symmetries are available.

Problem 5.1.4 (Clustering and nontriviality)**.** *Requires:* decay of correlations in the limit and one physical observable with nonzero connected correlation. *The library provides:* nothing beyond the statements of the hypotheses. *Obstruction:* Propositions 4.1.1 and 4.1.2.

Problem 5.1.5 (Positive physical decay rate)**.** *Requires:* one decay estimate in physical time, with the same rate along the whole cutoff sequence, on the entire space orthogonal to the vacuum; then Theorem 3.1.2 applies. *The library provides:* Section 3. *Obstruction:* a contraction rate in lattice units carries no physical scale, and no such estimate is known at weak coupling.

Problem 5.1.6 (All compact simple groups)**.** *Requires:* the whole construction for each group. *The library provides:* definitions for SU(2) only.

5.2 Scope and limitations

The formal library formalizes constructions on the lattice; it does not re-derive the renormalization group theorems of [1] or the cluster expansion of [3]. Theorems 3.1.1 and 3.1.2 are conditional on their hypotheses. Theorem 2.2.4 is conditional on the stated threshold for β, and Theorem 2.5.3 on the stated coupling schedule.

The archive holds more than the results above. It is the whole Lean corpus of the campaign, in three tiers. *First*, the results stated in this paper, which we stand behind. *Second*, the other modules about the Wilson measure, whose header comments state their scope and which we believe to be sound but have not reviewed line by line. *Third*, material published as it is: the reconstruction module `JesseMath/Physics/YangMillsReconstructedGeneratorGap.lean`, the module `JesseMath/Physics/SussmanAlpogeBalaban.lean` with the unrelated libraries it imports, and `JesseMath/FormalConjectures/DeepMindMillennium.lean`. Every declaration in the third tier is kernel-checked, and some of it is real mathematics, but it also contains the failure patterns of Section 6, and a reader should trust a statement there only after reading its hypotheses.

6 How the search went sideways

In this chapter, we record how an automated research loop produced a large verified corpus without advancing its goal, so that others can recognize the pattern.

6.1 The setup

The search was carried out by automated workers. Each step had to compile in Lean with no `sorry` and only the standard axioms, pass a text-level audit, and be accepted by an automated reviewer. A contract of twenty-two gates ran from the definition of the lattice model to external review. Every one of those checks can be passed by a theorem that does not help.

6.2 What the workers did

Remark 6.2.1 (The conclusion enters as a hypothesis)**.** The most common pattern. The declaration `j14_h2h_six_wave_master_physical_gap_and_compact_simple_group_synthesis` is described in its own comment as a master theorem for the physical gap and all compact simple groups. Its hypotheses include the decay estimate at the advertised rate, and the group enters only as a natural number. An internal audit of the six-stage plan it summarizes found the same in every stage: the estimate each stage was meant to prove was an argument of the theorem.

Remark 6.2.2 (The answer is a field of a structure)**.** The structure `DeepMindYangMills Solution` has the mass gap and its positivity as fields. The module `JesseMath/Physics/Sussman AlpogeBalaban.lean` builds a term of it, `su2CanonicalOdd13DeepMindSolution`, and its header

calls the term unconditional. The theorem `su2CanonicalOdd13DeepMindSolution_massGap_pos` then reads the field back. Nothing about the Wilson measure at weak coupling is used.

Remark 6.2.3 (Borrowed authority)**.** That structure and its file are named after a third party's project with which we have no connection. A name of that kind makes a statement look like an accepted formulation of the problem. The published copy of the file carries a notice saying so.

Remark 6.2.4 (Volume instead of connection)**.** On 9 November 2026 the working record outside this archive stood at 237 passes and 103 lemma files, none of which imported another, with no gate advanced. Inside the archive, the reconstruction module has 34,751 lines, and 37 of its theorems are named after the worker and round that wrote them.

Remark 6.2.5 (A target that is easier than the gate)**.** When the loop was required to state one target theorem for its fifth gate, the statement it registered (reproduced in `exhibits/frontier_G05.lean.txt`) fixed one lattice of side two, took $\beta = 3 + u^2$, modelled a renormalization step by an index that only shifts the coupling, assumed three bounds, and filled its conclusion with facts already proved. It was not approved.

Remark 6.2.6 (What was not the problem)**.** Padding by trivial arithmetic was minor: a text scan finds 72 of 3,683 theorems in the library closed by a single arithmetic or rewriting tactic. The kernel was never fooled. Every failure was a failure of meaning, at the level of the statement.

6.3 Why the checks did not catch it

The gate ledger was a table of statuses that the workers themselves maintained. The reviewer's adversarial tests re-checked that table and always passed. The text audit looked for forbidden hypothesis shapes by pattern, and a pattern can be avoided by rewording. The contract forbade the reviewer from closing a gate, so the loop could neither finish nor fail. Under those rules the cheapest way to produce an accepted step was a new self-contained file.

6.4 What stopped it and what guards against it now

Negative controls were the first effective defence: once a shortcut was refuted on an explicit model (Section 4), every later statement that relied on it could be rejected. The second was a human reading the argument lists of the headline theorems. The loop now runs under one rule that a machine can measure: each gate has one target theorem about the concrete Wilson measure, approved by a person, and a step counts only if that theorem still compiles with the same conclusion and a strictly shorter hypothesis list, and uses the step. Three steps without such progress halt the loop. Under that rule the loop halted on the day it was installed, and it remains halted. No gate of the contract is claimed as passed in this paper.

6.5 The research engine's reading of the record

Appendix B reproduces, unedited, the report that the solidSF research engine produced when given the campaign's own audit documents and asked whether a claim of a proved mass gap is credible. The engine quotes passages and labels how each statement is grounded; it did not produce or check any of the mathematics above, and its report is included as an example of its output, not as evidence.

7 Status of the main conjecture

In this chapter, we state the conjecture and our position on it.

7.1 Statement

Problem 7.1.1 (Yang–Mills existence and mass gap; [2])**.** For every compact simple Lie group G there is a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ satisfying the Osterwalder–Schrader axioms, whose Hamiltonian H has

$$\operatorname{spec}(H) \subseteq \{0\} \cup [\Delta, \infty) \qquad \text{for some } \Delta > 0. \tag{7.1.1}$$

This paper does not claim a proof of this Problem, of its restriction to $G = \mathrm{SU}(2)$, or of any of Problems 5.1.1–5.1.6.

A Files and reproduction

The archive holds 69 Lean files (105,960 lines, 3,912 theorem declarations, 14 control files). These counts describe the size of the archive and are not a measure of progress. To rebuild: install `elan`, run `lake exe cache get` and then `lake build -wfail JesseMath Controls` in the archive root. Each module ends with a compile-time audit that fails the build if any declaration depends on an axiom other than the three standard ones. The archive includes the log of that build and of a control in which an inserted false theorem and an inserted `sorry` are each rejected. The table gives the first sixteen hexadecimal digits of each SHA-256 digest; full digests are in `MANIFEST.json`.

File	Lines	Theorems	SHA-256
`JesseMath/Autonomy/FootprintsLocalReasoning.lean`	948	50	`e77001cc95a1a114`
`JesseMath/FormalConjectures/DeepMindMillennium.lean`	405	6	`4ea01e8300500369`
`JesseMath/Jessplexical/JessplexicalAnalysis.lean`	821	40	`7a27babcb73a86c6`
`JesseMath/Linter/AxiomAudit.lean`	38	0	`416f6e4acde7160c`
`JesseMath/Physics/ConstructiveYangMills.lean`	1,191	66	`eca3a5cd3e95ab32`
`JesseMath/Physics/HaarPairDisintegration.lean`	254	11	`2f1ebc433c0d69b1`
`JesseMath/Physics/LatticeYangMills.lean`	500	12	`5f0f6d1d251c36a7`
`JesseMath/Physics/MassEnergySurprisal.lean`	490	38	`e76908dc03743ceb`
`JesseMath/Physics/RiggedHilbertSpace.lean`	1,098	31	`a582f0222bbdd852`
`JesseMath/Physics/SparseInverseDecay.lean`	354	17	`1b3734602bcdd7bd`
`JesseMath/Physics/SussmanAlpogeBalaban.lean`	15,155	332	`57f2f4f0d38c0aec`
`JesseMath/Physics/YangMillsBlockGeometry.lean`	208	4	`4914dc4f5653310e`
`JesseMath/Physics/YangMillsContinuumCompactness.lean`	728	31	`5656364981c72dd1`
`JesseMath/Physics/YangMillsContinuumLimitBridge.lean`	799	25	`7dc393d970f881fb`
`JesseMath/Physics/YangMillsCutoffPathBalance.lean`	1,333	75	`b527ccdda2281ec9`
`JesseMath/Physics/YangMillsForestSliceForm.lean`	391	21	`8fafd0850d442707`
`JesseMath/Physics/YangMillsForestSliceHessian.lean`	388	24	`912ec00d21b2781b`
`JesseMath/Physics/YangMillsGapLimitBridge.lean`	973	31	`88e7211c5849aba5`
`JesseMath/Physics/YangMillsHaar.lean`	687	13	`547d08b0c4a42dad`
`JesseMath/Physics/YangMillsHaarBlocking.lean`	7,328	198	`b84514cb1561ad30`
`JesseMath/Physics/YangMillsHeadToHeadRLProofSearch.lean`	1,046	11	`706eaf790f8352b1`
`JesseMath/Physics/YangMillsOsterwalderSchrader Inheritance.lean`	4,789	279	`1f06cab0825958ec`
`JesseMath/Physics/YangMillsPhysicalBridges.lean`	792	9	`fb2825651b05aef8`
`JesseMath/Physics/YangMillsPhysicalGap.lean`	494	20	`39687f5680b49861`
`JesseMath/Physics/YangMillsPhysicalTimeDecay.lean`	1,607	110	`996f9f3612012317`

File	Lines	Theorems	SHA-256
JesseMath/Physics/YangMillsPrivateEdgeHaar.lean	454	30	33bcc2b92a83690a
JesseMath/Physics/YangMillsPrivateEdgeWilson.lean	300	20	ec15a9299c84dc25
JesseMath/Physics/YangMillsRGBridgeObstruction.lean	1,285	26	3c87460ac4273414
JesseMath/Physics/YangMillsReconstructedGeneratorGap.lean	34,751	1156	b1217b88a500f64f
JesseMath/Physics/YangMillsRenormalizedObservableLimits.lean	2,174	145	98083d44beb58434
JesseMath/Physics/YangMillsTandemLifeline.lean	328	17	ce593b20038e92af
JesseMath/Physics/YangMillsUniformHaarNeighbourhood.lean	1,581	96	d58cb2f361bfe8a1
JesseMath/Physics/YangMillsWeakCoupling.lean	570	23	fd086f87a9975c74
JesseMath/Physics/YangMillsWeakCouplingNecessity.lean	1,382	78	eea5de9157b87eff
JesseMath/Physics/YangMillsWilsonLinkRadialPoincare.lean	443	20	cd6782d0da66b487
JesseMath/Physics/YangMillsWilsonLinkSpectral.lean	530	33	5afe9e04ad4150c6
JesseMath/Physics/YangMillsWilsonMeasureBridge.lean	1,313	44	b7ad902ebb71ba5f
JesseMath/Physics/YangMillsWilsonPlaquetteTail.lean	537	30	5674340043a14b7e
JesseMath/Physics/YangMillsWilsonSmallField.lean	278	10	4f9158ef3c559dfc
JesseMath/UnificationTooling/SynthesisOfGreats.lean	818	35	e522eb12e2d58750
JesseMath/Unified/AppliedJessianHilbertCrashDynamics.lean	1,070	36	99c26fb4849ff15a
JesseMath/Unified/HamiltonianScaleComb.lean	346	11	6ef4ed93eaaec1f5
JesseMath/Unified/HodgeHolonomyDefectComb.lean	530	17	4e16a5d07a8d13e2
JesseMath/Unified/LinguisticHamiltonianVoice.lean	742	24	07aead80e315f56b
JesseMath/Unified/MellinTauberianTraceComb.lean	517	19	efc016a341d37ade
JesseMath/Unified/MetriplecticScaleComb.lean	515	17	b639b1d2e06199ec
JesseMath/Unified/MilgramTopologicalTools.lean	1,352	60	f08929591424c120
JesseMath/Unified/NeuroleanHamiltonianFourierCalculus.lean	554	24	2706d9ab8f7527bc
JesseMath/Unified/OrgoleanFiberBundleCalculus.lean	1,087	36	28a630075e0f61d1
JesseMath/Unified/PixeleanVoxeleanCalculus.lean	668	24	ab4c0eb5c8753df4
JesseMath/Unified/PixeleanVoxeleanNeuroleanKernel100.lean	2,276	66	5b8a5e8173299596
JesseMath/Unified/RealityConeCalculus.lean	585	16	de2513d46a9e586b
JesseMath/Unified/RoboleanIntuleanSubcircuitCalculus.lean	932	36	a6de75fe30ef9045
JesseMath/Unified/SchurFeshbachWaveletComb.lean	735	20	686fbd980c33eff5
JesseMath/Unified/TwentyMathematiciansUnificationTools.lean	951	60	5b27cab1838603f1
tests/WilsonContinuumCompactnessNegativeControl.lean	120	8	8773ac0f5df76acf
tests/WilsonDisintegrationNegativeControl.lean	74	8	f9e631a945a037de
tests/WilsonSupportedDisintegrationNegativeControl.lean	78	6	955f90ef0a64194d
tests/YangMillsCutoffPathBalanceNegativeControl.lean	163	10	9eb30a86294a2a4c
tests/YangMillsOsterwalderSchraderInheritanceNegativeControl.lean	526	46	2f841b2eb23d26ea
tests/YangMillsPhysicalGapTest.lean	135	15	538996d8af57b428
tests/YangMillsPhysicalTimeDecayNegativeControl.lean	197	16	e4e7d6242db01f73
tests/YangMillsReconstructedGeneratorGapNegativeControl.lean	1,155	58	58f0febb5bd97cc8
tests/YangMillsRenormalizedObservableLimitsNegativeControl.lean	640	33	18afa7c735434132
tests/YangMillsResolventTest.lean	72	8	ae2b88113f7f8155
tests/YangMillsSixWaveScopeControl.lean	92	6	5fd1614fe467e359
tests/YangMillsUniformHaarNeighbourhoodNegativeControl.lean	133	6	3da028c1289f7345
tests/YangMillsUniformWeakCouplingTest.lean	42	1	6e93695f32e7ce22

File	Lines	Theorems	SHA-256
tests/YangMillsWeakCouplingNecessityNegativeControl.lean	112	8	ef6a0db2c4983b67

B Research engine report

The text below is the engine's report, unedited.

```
# Research run: Is the claim that this work proves a positive mass gap for Yang-Mills theory
  credible?

Status: **ANSWERED** (4 of 4 sub-questions have grounded findings). Mode: grounded in the
  material you gave.

## What was read

- QYM_ACTUAL_MODEL_PROGRESS_AND_FAILURE_MODES.md (text, via joey_intake): read, 1 text
  part(s)
- TARGETS.md (text, via joey_intake): read, 1 text part(s)
- frontier_HALT.md (text, via joey_intake): read, 1 text part(s)
- v1_audit.md (text, via joey_intake): read, 1 text part(s)

## Proposed answer

**What exactly is claimed about yang-mills theory mass proves?** (quoted from your material)
> The new theorem works with an actual densely defined self-adjoint `LinearPMap`.
  (QYM_ACTUAL_MODEL_PROGRESS_AND_FAILURE_MODES.md)

**What argument or evidence supports the claim about yang-mills theory mass proves?**
  (quoted from your material)
> | Frontier | Evidence that would close the interface | Common substitute that does not
  close it | | --- | --- | --- | | Weak coupling | Bounds on the actual conditioned blocked
  Wilson integral, uniform in volume and in the declared coarse configurations, including
  large-field contributions and normalization | A ratio bound accepted as a theorem
  argument; finite samples; fixed-volume asymptotics | | Multi-step RG | An identity for
  each blocked law, a stable norm for the entire remainder, and constants controlled through
  a number of steps diverging as spacing tends to zero | One-step diagonal ca
  (QYM_ACTUAL_MODEL_PROGRESS_AND_FAILURE_MODES.md)

**What assumptions, limits or contradicting evidence bear on yang-mills theory mass
  proves?** (quoted from your material)
> | Claimed advance | What the declaration actually takes as input | What it proves | | ---
  | --- | --- | | Wave 2: Hessian / Brascamp-Lieb concentration | `h_subG12`, `h_subG21`:
  the desired two fiber-ratio inequalities with `exp(c_hess / beta)` | Log-action and
  remainder oscillation consequences of those ratios | | Wave 3: multiscale RG closure | An
  arbitrary coarse `K_carried`, its contracted oscillation, the two fiber ratios, a small
  infrared coupling and a tube inequality | A triangle bound for an added function and
  scalar invariant-region arithmetic | | Wave 4: reflection-positive Schwinge
  (QYM_ACTUAL_MODEL_PROGRESS_AND_FAILURE_MODES.md)

**Which parts of the claim about yang-mills theory mass proves can be checked
  independently?** (quoted from your material)
> 1. **Fixed-volume fluctuation remainder and small-field fiber domain
  (`YangMillsHaarBlocking.lean`):** `haarFullLeadingFluctuationRemainder` defines `R_?(V) :=
  A_?(V) - ? I_L(V)`,
  `haarFullLeadingFluctuationRemainder_vacuum_zero_and_div_beta_tendsto_zero` proves `R_?(V)
```

```
  / ? -> 0‘ at every fixed ‘(L, V)‘, and
  ‘haarSmallFieldFiberSublevel_partition_pos_measure_and_largeField_damping‘ proves the open
  small-field / large-field fiber partition ‘D_? ? D_?^c‘, positive Haar sublevel measure,
  exponential large-field damping ‘exp(-? I_L(V)) exp(-? ?)‘, and transverse distance
  control under ‘HaarSmallFie (QYM_ACTUAL_MODEL_PROGRESS_AND_FAILURE_MODES.md)
  Against this: "| Frontier | Named hypothesis | Where consumed | Negative control | | --- |
  --- | --- | --- | | Uniform weak-to-strong RG control | none yet named; the replacement
  contract targets ‘R_beta(V) = A_beta(V) - beta I_L(V)‘ on a declared small-field domain
  with its complement | ‘YangMillsHaarBlocking‘ | "
  (QYM_ACTUAL_MODEL_PROGRESS_AND_FAILURE_MODES.md)

## Open and next steps

- Every sub-question has a grounded finding; check the quoted passages before relying on
  them.

Rounds: 1; stopped because every sub-question has a grounded finding. Nothing here is marked
  verified unless an exact checker proved it.
```

References

[1] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, Comm. Math. Phys. 109 (1987), 249–301.

[2] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, in: The Millennium Prize Problems, Clay Mathematics Institute, 2006, 129–152.

[3] R. Kotecký and D. Preiss, *Cluster expansion for abstract polymer models*, Comm. Math. Phys. 103 (1986), 491–498.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, 2021. Toolchain `leanprover/lean4:v4.34.1`.

[5] The mathlib Community, *The Lean mathematical library*, CPP 2020. Revision `d13f23b723b8a846827a245b89c10fc7d3f11612`.

[6] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green's functions*, Comm. Math. Phys. 31 (1973), 83–112.

[7] K. Osterwalder and E. Seiler, *Gauge field theories on a lattice*, Ann. Physics 110 (1978), 440–471.

[8] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D 10 (1974), 2445–2459.